Multilayer Perceptron for Classification of Mobile Price Range using Kaggle Dataset

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Group Members

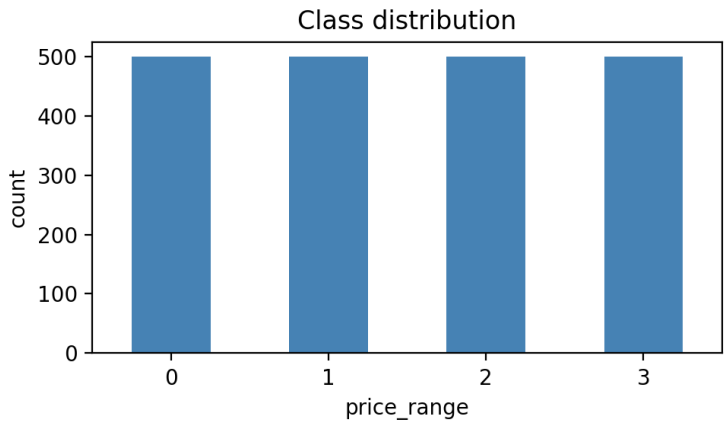
1. Yusuf Imantaka Bastari - 24/532731/PA/22530
2. **[TODO: member name - NIM]**
3. **[TODO: member name - NIM]**

Job Description for each member

Member	Job description
Yusuf Imantaka Bastari	**[TODO]**
[TODO]	**[TODO]**
[TODO]	**[TODO]**

Problem Case

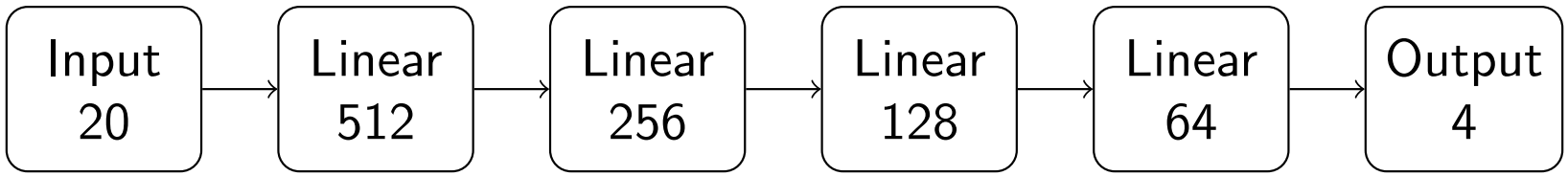
- Kaggle *Mobile Price Classification* dataset
- Predict the phone `price_range` (0–3) from its specifications
- 2000 samples, 20 numerical features (e.g. `ram`, `battery_power`, `px_height`)
- No missing values, no duplicates
- 4 balanced classes (500 samples each)
- Main metric: **macro F1-score**



Solution Methodology

1. Standardize the features (fit on the training folds only, to avoid leakage)
2. Evaluate with **stratified 5-fold cross-validation**
3. Train a Multilayer Perceptron (MLP) classifier in PyTorch
4. Ablation study on network layers: dropout, activation function, batch normalization
5. Ablation study on loss functions
6. Compare all experiments using the mean macro F1-score over the 5 folds

Neural Network Architecture



- Fully connected MLP with 4 hidden layers:
  $20 \to 512 \to 256 \to 128 \to 64 \to 4$
- Activation after every hidden layer (baseline: ReLU)
- Optional batch normalization (after linear) and dropout (after activation)
- Output: 4 logits, one per price range

Learning Strategy and Parameter Tuning

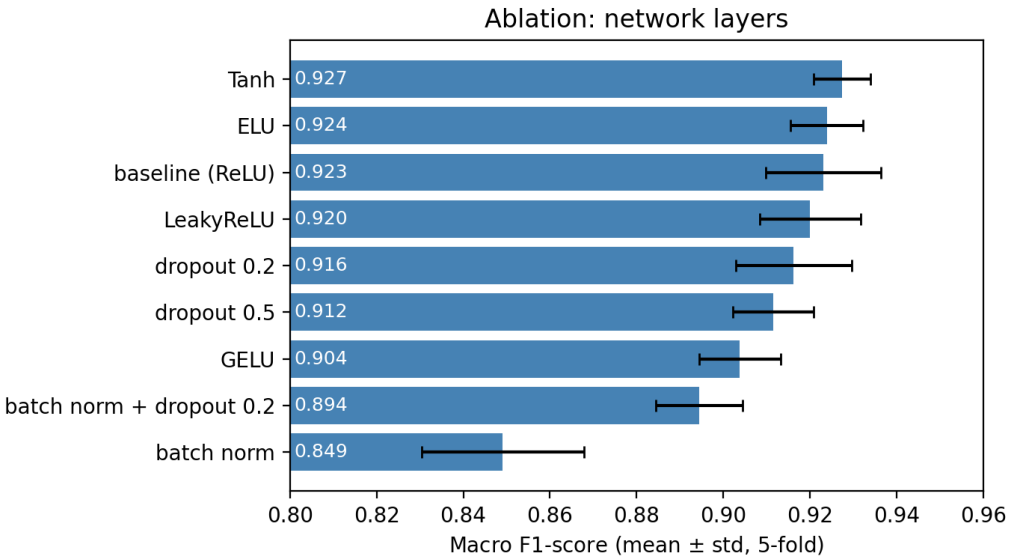
Fixed training setup

- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 64
- Epochs: 20
- Seed: 42 (same for every experiment)
- Stratified 5-fold, shuffled

Tuned by ablation

- Layers: dropout (0.2, 0.5); ReLU, LeakyReLU, ELU, GELU, Tanh; batch norm
- Loss: cross-entropy, label smoothing (0.1), focal ($\gamma = 2$), multi-margin, MSE on one-hot

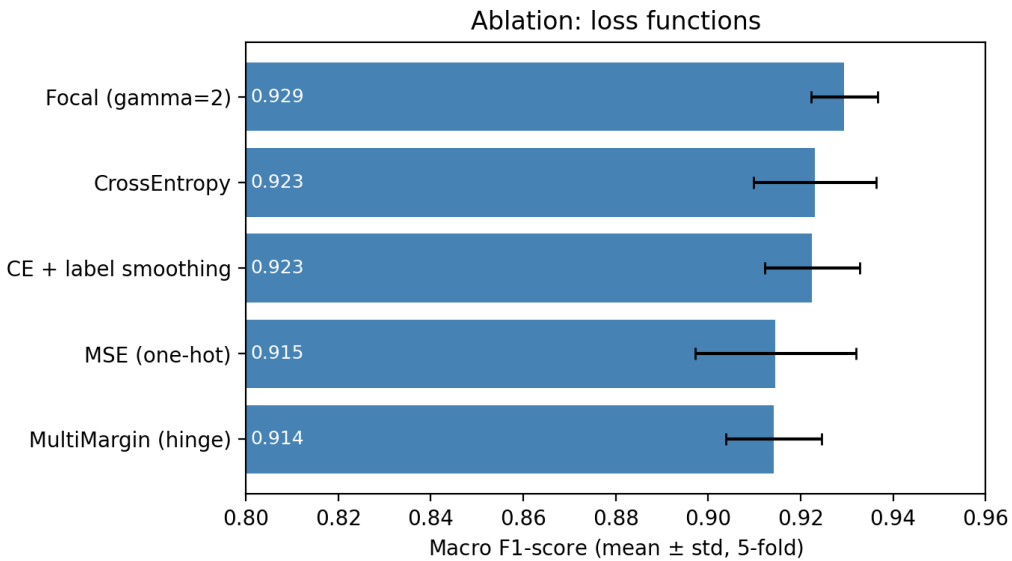
Results Experiment: Network Layers



Loss fixed to cross-entropy.

- Best: Tanh (0.927)
- ELU, ReLU, LeakyReLU are within one std of Tanh
- Dropout does not help
- Batch norm hurts the most (0.849)

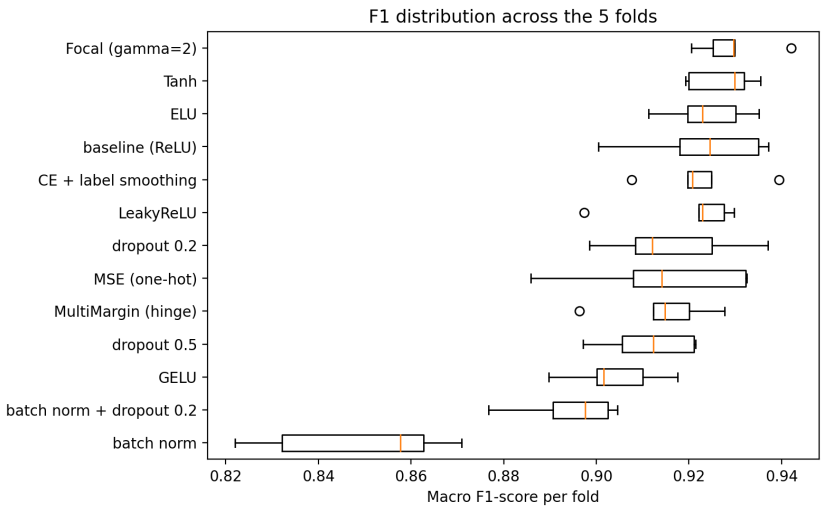
Results Experiment: Loss Functions



Architecture fixed to baseline (ReLU, no dropout).

- Best: focal loss (0.930)
- Cross-entropy (0.923) and label smoothing (0.923) are similar
- Multi-margin and MSE are lower (0.914–0.915)

Results Experiment: Summary



Experiment	F1 (mean ± std)
Baseline (ReLU, CE)	0.923 ± 0.013
Tanh	0.927 ± 0.007
ELU	0.924 ± 0.008
Dropout 0.2	0.916 ± 0.013
Batch norm	0.849 ± 0.019
Focal loss	**0.930** ± 0.007
Label smoothing	0.923 ± 0.010
Multi-margin	0.914 ± 0.010
MSE (one-hot)	0.915 ± 0.017

Discussion

- Most variants are within about one standard deviation of the baseline, so the differences among the top models are not clearly significant
- Dropout did not improve F1; with only 20 epochs the baseline is likely not overfitting enough to need extra regularization
- Batch norm clearly lowered F1 (0.849); adding dropout partly recovered it (0.894)
- Smooth activations (Tanh, ELU) are slightly better and more stable across folds than GELU
- Focal loss gave the highest mean F1 with a low std, but its gain over cross-entropy is small (about 0.007)

Conclusion

- An MLP reaches a macro F1-score of about 0.92–0.93 on mobile price range classification
- Best loss: focal loss (F1 = 0.930); best layer variant: Tanh activation (F1 = 0.927)
- Batch norm and, in this setup, dropout did not help
- Future work: combine the best layer and loss settings, tune the learning rate and number of epochs, and confirm on a held-out set

GitHub Link

https://github.com/yusufimantaka/mobile-price-range-classification

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